

Bayesian Methods for Clinical Trials

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Practical III: Borrowing of information in basket trials

Recall the BRAF-V600 basket trial in Lecture 9. It is a phase II PoC trial to simultaneously evaluate the efficacy of vemurafenib, a novel anti-cancer treatment, in several patient subpopulations. The primary endpoint is binary: response or non-response.

Open the R script titled `BasketTrials.R` and ensure that the JAGS model files, `StandardHM.txt`, `EXNEX.txt` and `Stand-alone.txt`, are in the working directory (Use `setwd()` to modify if necessary). For each substudy, the Bayesian analysis model returns the posterior mean of response rate together with a credible interval.

- (a) Read the `StandardHM.txt` and `EXNEX.txt` scripts to understand how these Bayesian models are fitted for borrowing of information.
- (b) Use the code written in `BasketTrials.R` to run an analysis for the BRAF-V600 trial.

```
# BRAF-V600 basket trial data
> nMod = 6
> r = c(8, 1, 6, 0, 2, 6)
> n = c(20, 8, 18, 10, 7, 32)
```

Set $\mu \sim N(0, 10^2)$, $\tau^2 \sim HN(0.125)$ to implement the standard hierarchical model and, additionally, the prior mixture weight as (0.5, 0.5) to implement the EXNEX model. For comparison, implement the stand-alone analysis model.

What are the posterior mean for the response rate, denoted by p_k , per substudy $k = 1, \dots, 7$?
What are the posterior probabilities that $\Pr(p_k > 0.25 \mid \text{data})$?

- (c) Install and load the ‘bmabasket’ package in R. Using the `bma()` function, run an analysis for the BRAF-V600 trial using Bayesian model averaging. What are the posterior means of the response rates per substudy? What are the posterior probabilities that $\mathbb{P}(p_k > 0.25 \mid \text{data})$?
- (d) Retain the prior specification unchanged from (b). Analyse the hypothetical datasets below using the standard hierarchical modelling approach, EXNEX model, BMA and stand-alone analysis. Again, what are the posterior mean for the response rate, denoted by p_k , per substudy $k = 1, \dots, 5$? What are the posterior probabilities that $\Pr(p_k > 0.25 \mid \text{data})$?

Hypothetical dataset of a consistency scenario:

	Substudy 1	Substudy 2	Substudy 3	Substudy 4	Substudy 5
Number of patients	10	10	20	20	17
Number of responses	5	4	12	7	9

Hypothetical dataset of an inconsistency scenario:

	Substudy 1	Substudy 2	Substudy 3	Substudy 4	Substudy 5
Number of patients	10	10	20	20	17
Number of responses	2	7	15	4	5

- (e) Compare the resulting posterior mean estimate and posterior standard deviation under both scenarios. What do you notice about the relative advantages and disadvantages of each analysis model?
- (f) Set the argument of `Prior.tau.HN` to a larger value, for example, 0.25. What impact does this change have on the posterior means and credible intervals?
- (g) When using the EXNEX for analysis, set the argument of `pMix` to a range of large values, for example, 0.8, for the probability of exchangeability. How does this quantity affect the results?
- (h) How would you expect the subtrial-wise type I error rate and statistical power to change using analysis models that permit borrowing of information? Perform a small-scale simulation study to develop intuition.
- (i) Do you have any suggestions on how to better control for a type I error?